

Gold's ecommerce case study

Brief Introduction to the company

Gold is a fictional e-commerce company specializing in adventure and outdoor sports gear. With a catalog of nearly 300 products, Gold has steadily grown its customer base across the United States, Canada, and Australia. As the company prepares for international expansion, leadership wants a deeper understanding of how the business has evolved over time—what drives sales, how customer behavior has shifted, and which products are contributing most to growth.

Gold has provided **five years of historical transactional data**, beginning from late **2010**, followed by complete annual data for **2011, 2012, 2013, and 2014**, including the first month of each year. This dataset captures order activity, customer demographics, and product performance, offering a valuable opportunity to analyze trends and identify patterns that can guide strategic expansion decisions.

Objective of the analysis

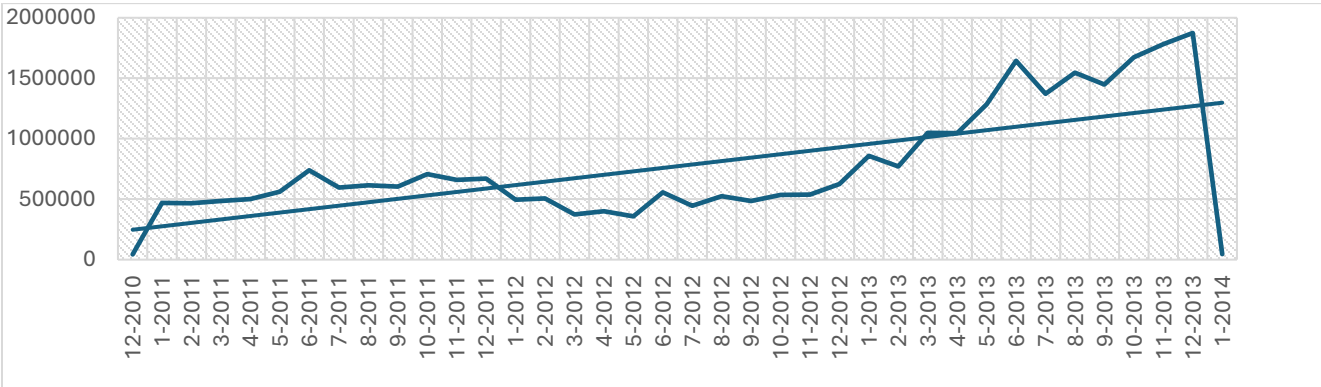
The objective of this analysis is to examine Gold's performance over a five-year period and uncover the key factors influencing revenue, customer growth, and product success. Specifically, the project aims to:

- Analyze how sales, orders, and customer counts have changed over time
- Evaluate cumulative performance to understand long-term growth
- Assess product categories and individual product contributions
- Identify regional trends across the US, Canada, and Australia
- Segment customers based on purchasing behavior
- Provide data-driven recommendations that support Gold's expansion into new markets

Tools used

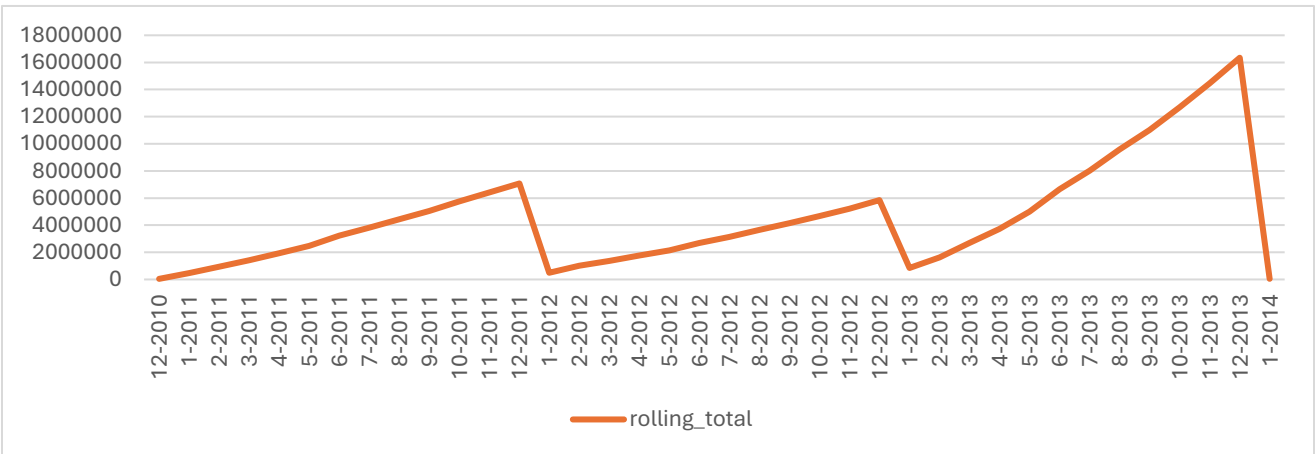
SQL used for analysis and Excel is used for visualization for the data.

Changes Over Time Analysis



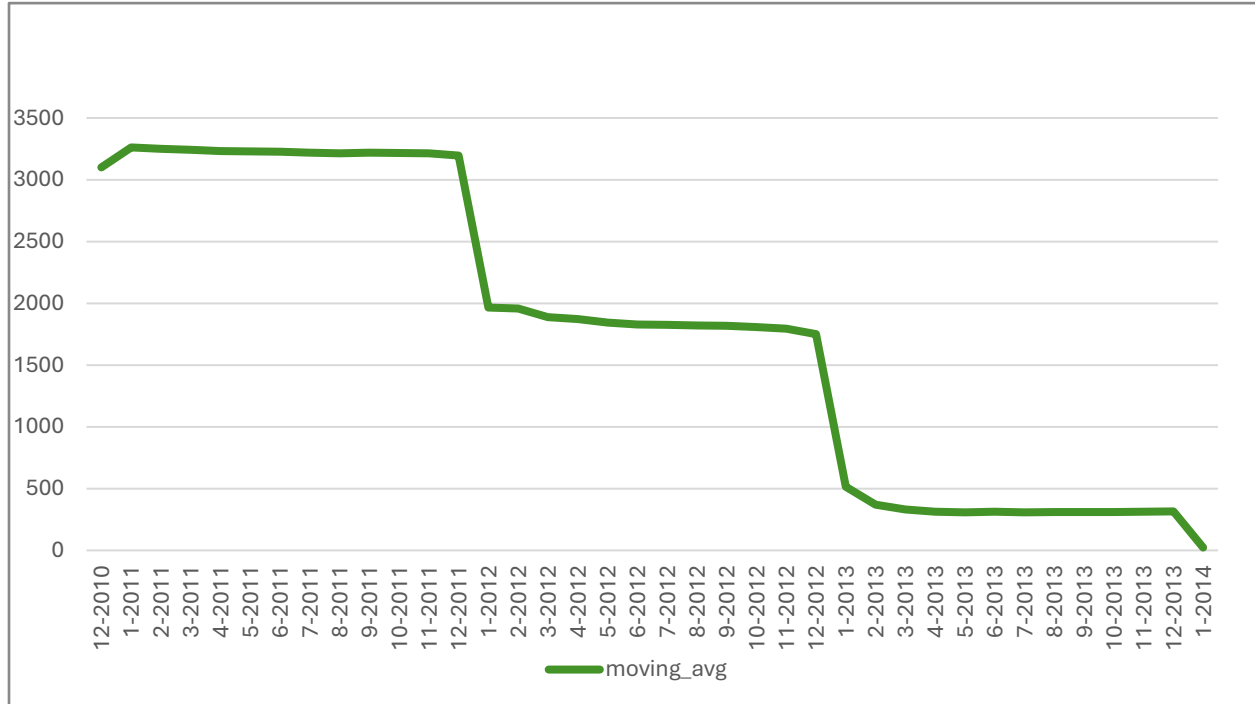
The revenue trend shows a clear upward trajectory from the company’s start in late 2010 through 2013, reflecting consistent business growth over the five-year period. December stands out as a recurring peak, which aligns with the holiday season and higher consumer spending on adventure and outdoor products. In 2012, revenue experienced a noticeable dip—likely influenced by seasonal fluctuations or market conditions—but the business recovered strongly in 2013. Growth accelerated significantly throughout 2013, reaching its highest point in December 2013. However, this surge is followed by a sharp drop at the beginning of 2014, suggesting either incomplete data for the year or an operational or market-driven slowdown.

Rolling total analysis



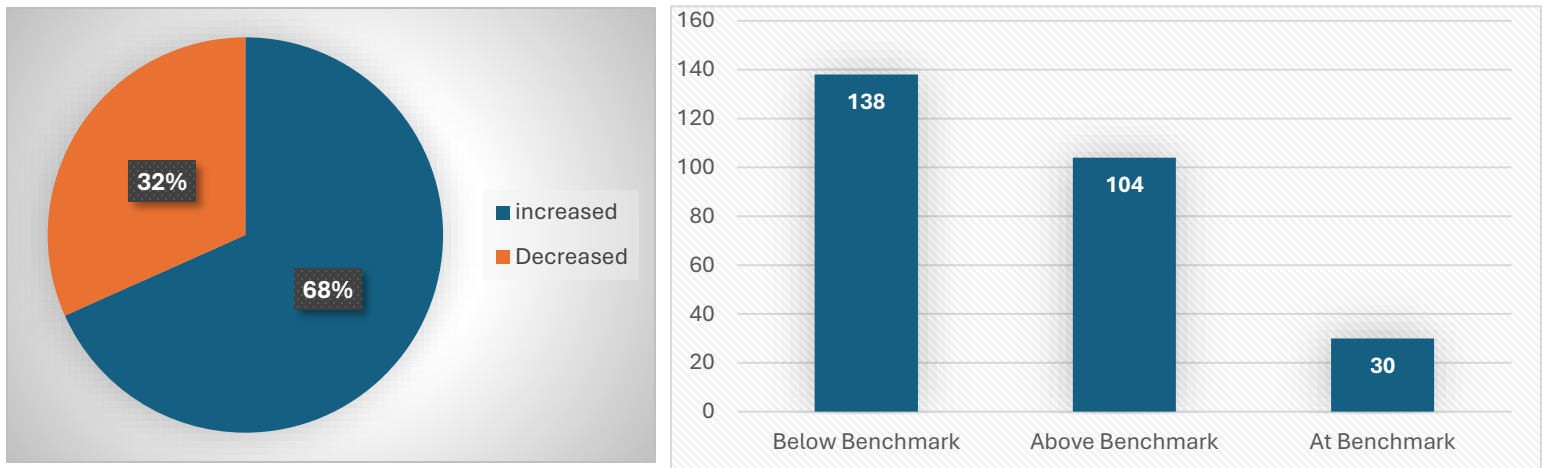
The year-partitioned rolling total reveals a clear pattern of accelerating revenue performance for Gold’s Ecommerce. Each annual cycle begins with a reset, yet the pace at which revenue accumulates grows progressively stronger from 2010 to 2013. The curves show smoother and faster compounding, indicating an expanding customer base and improved market penetration. The dramatic surge in 2013, far steeper than previous years, suggests a breakthrough year driven by increased product adoption, effective promotions, or broader regional reach.

Moving average analysis



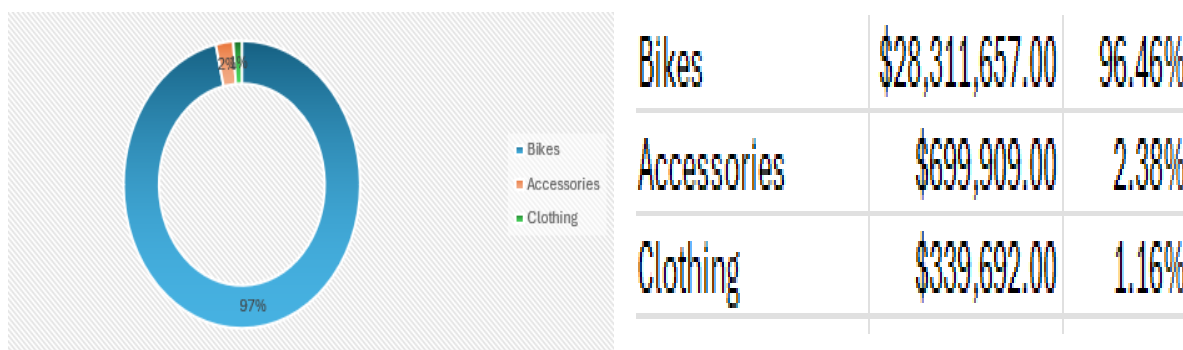
The moving average chart shows a steady and pronounced decline in average product prices year after year. In 2010 and 2011, average prices remained relatively stable at higher levels. However, there is a significant drop at the start of 2012, followed by another steep decline entering 2013. By 2014, the moving average reaches its lowest point. This trend suggests that Gold gradually reduced prices, likely due to increased competition, expansion into lower-priced product categories, or deliberate pricing strategies to boost sales volume. The consistent downward movement indicates that products became more affordable over time, potentially supporting wider customer adoption—even if it reduced average revenue per unit.

Year on year Analysis



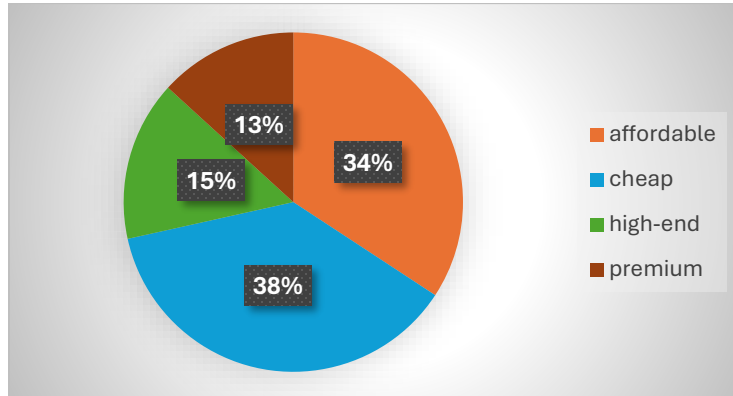
The year-on-year comparison highlights strong revenue momentum across Gold's product portfolio. As shown in the pie chart, **nearly 70% of product-year combinations experienced an increase in revenue** compared to the previous year. This indicates healthy demand growth and consistent performance across a majority of products. The remaining 32% that showed declines should be examined closely to identify potential issues such as price changes, reduced marketing exposure, or product lifecycle effects. The benchmark analysis reveals that many products performed **below their long-term average revenue**, suggesting room for improvement in consistency. However, **104 instances exceeded the benchmark**, demonstrating strong competitive performance and the potential for scaling these high-performing items.

Part to whole analysis



In this segment we analyze which category holds the most impact for the business. Bikes bring most of the business as we can infer from the graph that almost all of the business comes from the bike category. Gold gets negligible business from clothing and slight business from accessories. However it is not wise to depend on only one source of income as they say one must always diversify. Hence it is wise for gold to focus on accessories and clothing segment.

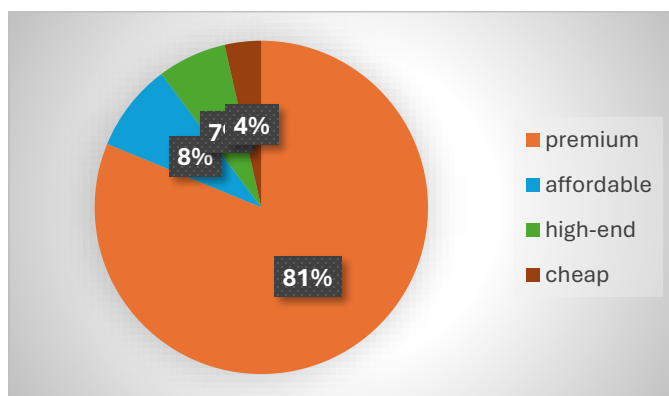
Product segment distribution



The chart indicates that Gold's product strategy is strongly centered around affordability. A large portion of the catalog falls within the budget (below £100) and mid-range (£100–£500) segments, reflecting a focus on accessible adventure gear for the broader market. At the same time, the company caters to higher-value customers as well: roughly 15% of products fall into the high-end £500–£1,000 range, offering more advanced or specialized equipment. Additionally, about one-eighth of the catalog consists of premium products priced above £1,000, signaling an intentional offering for enthusiasts willing to invest in top-tier gear.

This distribution suggests that while acquisition is strong, customer retention and value expansion need improvement. Gold should focus on strengthening loyalty through enhanced customer support, personalized engagement, and structured rewards programs. Initiatives that encourage repeat purchases and higher order values can increase the share of VIP customers and build a more stable, long-term customer base.

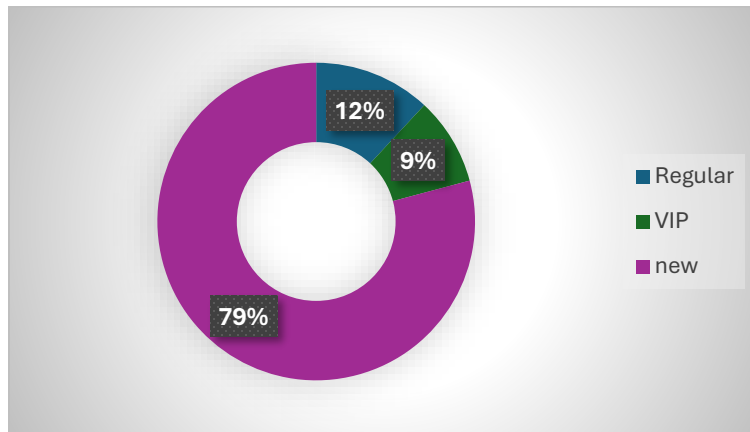
Revenue share by product segment



The chart highlights a striking imbalance in revenue contribution across product segments. Although the Premium category represents only a small share of the total product catalog, it contributes an overwhelming 81% of total revenue, making it the company's primary profit

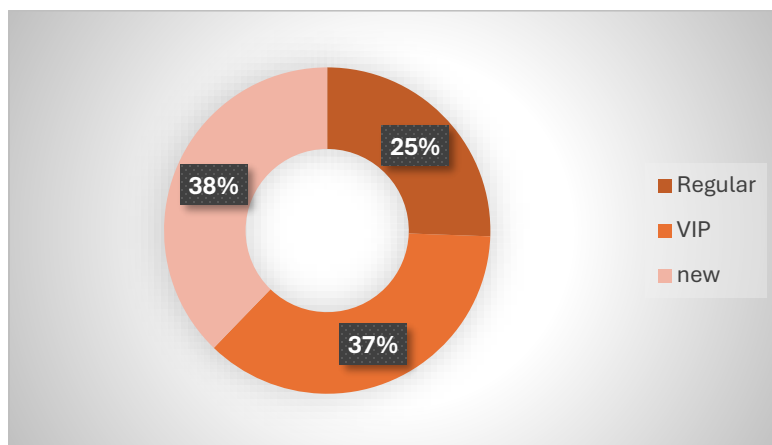
driver. In contrast, the affordable, high-end, and cheap segments collectively generate less than 20% of revenue despite accounting for the majority of product offerings.

Customer distribution by segment



The chart reveals that nearly 80% of customers are new, meaning they have been purchasing from the company for less than a year. This indicates that Gold excels at attracting fresh customers, reflecting strong marketing reach and brand visibility. However, only about 20% transition into long-term customers, categorized as either *Regular* or *VIP* based on annual spending. Customers with less than £5,000 in purchases are classified as Regular, while those exceeding £5,000 fall into the VIP group.

Revenue distribution by customer segment



Although new customers make up nearly 80% of the customer base, they contribute only about 37% of total revenue. In contrast, the loyal customer segments — Regular and VIP — represent

just 20% of the population but generate more than 63% of the revenue. This shows that long-term customers are significantly more valuable, contributing almost two-thirds of overall sales despite their smaller numbers.

Note of thanks to Data with Baraa Youtube channel

<https://www.youtube.com/@DataWithBaraa>